

The Brick-Ladder Theorem:
 $\tau^*(\delta) = 27/(4 - \delta^2)$ exactly
on 12/12 rungs $\delta_k = \pi/k$

$\kappa(\delta) = 2 - \delta^2/2$ · the clock is brick-invariant · the tick does not select the brick

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Series:	Monograph No. 02 of the “Twelve Theorems” cycle
Machine:	brick_scan_tick.py · SymPy, exact
Verdict:	einstein_direct/results/brick_scan_tick.json
Status:	PROVED EXACTLY (12/12 rungs of the ladder)

Abstract

This monograph records the brick-ladder theorem — the exact law of motion of the clock root along the spinor ladder of bricks $\delta_k = \pi/k$. On each of the twelve rungs (the baseline point plus $k = 2 \dots 14$) the greatest common divisor of the clock pair (the UV book, the mass-defect book) is computed by SymPy in closed form and has exactly one common positive root; the whole tuple of roots is described by the single formula $\tau^*(\delta) = 27/(4 - \delta^2) = 27k^2/(4k^2 - \pi^2)$, equivalent to the holonomy family $\tau^*(\kappa) = 27/(2\kappa)$ at $\kappa(\delta) = 2 - \delta^2/2$.

The complete numerical tuple of roots from 17.617134 ($\pi/2$) to 6.836058 ($\pi/14$) is given, together with the layers B1 (the brick independence of the flat line), B3 (the tick is non-monotone, not a power law, minimum at $k = 12$ — the brick is not selected), B4 (phantoms outside 1% of the ladder) and B5 (the κ -independent book defect), the corollaries, the honest caveats and the reproduction protocol.

Keywords: brick ladder, $\delta = \pi/k$, $\tau^*(\delta) = 27/(4 - \delta^2)$, clock invariance, tick, SymPy, machine proof.

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1. Introduction

The second monograph of the series records the brick-ladder theorem — the result of campaign v18a (BRICK-LADDER-TICK, machine `brick_scan_tick.py`, LEDGER §29). The campaign asked whether the exact structure of the clock pair (monograph No. 1) survives on the intermediate bricks — the spinor corrections $\delta_k = \pi/k$ of the ladder $k = 2 \dots 14$ — or whether the exactness of the baseline point was a lucky coincidence of two chosen values of the constant.

The machine answer is unambiguous: the structure survives on all twelve rungs (the baseline point plus eleven bricks). On each rung the greatest common divisor of the clock pair is written out by SymPy in closed form, the pair has exactly one common positive root, and the whole tuple of roots is described by one formula $\tau^*(\delta) = 27/(4 - \delta^2) = 27k^2/(4k^2 - \pi^2)$, equivalent to the holonomy family $\tau^*(\kappa) = 27/(2\kappa)$ under the convention $\kappa(\delta) = 2 - \delta^2/2$. The clock is thus invariant under the choice of brick: the parameter changes, the law does not.

The second part of the campaign honestly separates the clock from selection: the tick obstruction, measured on the same ladder, proved non-monotone in δ , not a power law, and reaches its minimum at $k = 12$, not at the surviving brick $k = 7$. The tick therefore does not select $\delta_C = \pi/7$ — a fact filed separately as part of the structural barrier (monograph No. 10). Monograph No. 2 records only the algebraic side: the exactness of the clock along the ladder and the brick independence of the flat line.

2. Notation and Setting

A brick is a spinor correction δ embedded into the constant of the corrected system. The convention of campaign v18a uses the additive form $\kappa(\delta) = 2 - \delta^2/2$: the baseline answers $\delta = 0$, the surviving brick $\delta_C = \pi/7$ (the sector $b_C = \pi^2/98$, where $2 \cdot \pi^2/98/2 = \pi^2/98$ gives $\kappa = 2 - \pi^2/98$). The ladder of the campaign: $\delta_k = \pi/k$ for $k = 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14$ plus the baseline rung $\delta = 0$ — twelve static points in all.

For each rung the machine assembles the corrected system (the same twelve tower equations and six chain equations with κ -dependent books), substitutes the chain forms and

computes the GCD of the clock pair (UV_ξ3, Mdef_ξ5). The theorem is posed as follows: to prove that on each rung the GCD has the form $T0h^2 \times$ (a quadratic clock factor), that the positive root is unique, and that the roots of all rungs are described by a single formula in δ . Additionally the brick independence of the flat line (layer B1) and the status of the tick as a non-selection (layer B3) are recorded.

Table 2.1. Notation of the monograph

Symbol	Meaning
$\delta_k = \pi/k$	brick of rung k ($k = 2 \dots 14$)
$\kappa(\delta) = 2 - \delta^2/2$	additive convention of the corrected-system constant
$\tau^*(\delta)$	the unique common positive root of the clock pair at rung δ
B1–B5	layers of the machine verdict v18a (flat line, clock, tick, phantoms, book defect)

3. Statement of the Theorem

Theorem 2 (the brick ladder of the clock). Let $\delta_k = \pi/k$, $k \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14\}$, and $\delta_0 = 0$; $\kappa(\delta) = 2 - \delta^2/2$. Then on each of the twelve rungs of the ladder:

(i) the greatest common divisor of the clock pair (UV_ξ3, Mdef_ξ5) on the static chain equals $T0h^2(\pi - k)(\pi + k) \cdot ((k^2 - \pi^2) \cdot 4T0h^2/k^2 - 27k^2) \cdot k^2/(k^2 - \pi^2)$ in the machine form — that is, $T0h^2$ times one quadratic clock factor; the baseline rung gives $T0h^2(4T0h^2 - 27)/4$;

(ii) the pair has exactly one common positive root

$$\tau^*(\delta_k) = 27/(4 - \delta_k^2) = 27k^2/(4k^2 - \pi^2) \quad (2.1)$$

and formula (2.1) is exact on all twelve rungs: the GCD vanishes at $\tau^*(\delta)$ (gcd_zero_at_tau_star = true) and uniqueness holds (unique_common_clock = true) on each rung;

(iii) under the convention $\kappa(\delta) = 2 - \delta^2/2$ formula (2.1) is equivalent to the holonomy family $\tau^*(\kappa) = 27/(2\kappa)$;

(iv) the flat line of equilibria (monograph No. 6) is brick-independent: on all twelve rungs it carries $F = r = V = 0$ with machine zero ($F \leq 10^{-7}$ in the float64 layer of the campaign).

4. Sketch of the Machine Proof

The proof is executed by the machine `brick_scan_tick.py` under a single procedure for each rung: (1) assembly of the corrected system at $\kappa(\delta_k)$; (2) reduction of the residues on the chain; (3) SymPy computation of the GCD of the pair in exact arithmetic over $\mathbb{Q}(\pi)$; (4) verification of the vanishing at the booking root, of the uniqueness of the positive root and of the coincidence of the individual roots of the two books. All twelve runs are deterministic (BLAS = 1, offline on the saved JSON dependencies) and reproduce with one command.

The key step is the recognition of the common structure. For rung k the machine returns a factorization of the form $T0h^2(-k^2 \cdot 4T0h^2 + \pi^2 T0h^2 + 27k^2)(\pi - k)(\pi + k) \times (\text{a rational factor})$: the trivial factor $T0h^2$, the spinor cofactors $(\pi - k)(\pi + k)$ originating from the denominator of the κ -correction, and the clock quadratic with a single positive root. The reduction of all factorizations to the single formula (2.1) is an algebraic identity verified by SymPy on each rung separately.

$$GCD_k = T0h^2(-k^2 \cdot 4T0h^2 + \pi^2 T0h^2 + 27k^2)(\pi - k)(\pi + k) \cdot k^2/(k^2 - \pi^2) \quad (2.2)$$

Layer B1 (the brick independence of the flat line) is verified by a march along the a-pure kernel direction on each rung: F , r and V remain machine zeros; in the float64 layer of the campaign an upper bound $F \leq 10^{-7}$ is registered on all rungs — a limit of the representation, not of the system.

5. Numerical Results: the root tuple of the ladder

Table 2.2. The exact tuple of roots $\tau^*(\delta_k)$ per the v18a JSON verdict (12/12 rungs)

Rung	τ^* (symbolic)	τ^* (numeric)	T0h root
$\delta = 0$ (baseline)	27/4	6.750000	2.59807621
$\pi/2$	$108/(16 - \pi^2)$	17.617134	4.19727696
$\pi/3$	$243/(36 - \pi^2)$	9.299515	3.04951058

Rung	τ^* (symbolic)	τ^* (numeric)	T0h root
$\pi/4$	$432/(64 - \pi^2)$	7.980729	2.82501835
$\pi/5$	$675/(100 - \pi^2)$	7.489149	2.73663103
$\pi/6$	$972/(144 - \pi^2)$	7.246680	2.69196575
$\pi/7$ (surviving)	$1323/(196 - \pi^2)$	7.107920	2.66606830
$\pi/8$	$1728/(256 - \pi^2)$	7.020669	2.64965448
$\pi/9$	$2187/(324 - \pi^2)$	6.962077	2.63857480
$\pi/10$	$2700/(400 - \pi^2)$	6.920763	2.63073430
$\pi/12$	$3888/(576 - \pi^2)$	6.867676	2.62062507
$\pi/14$	$5292/(784 - \pi^2)$	6.836058	2.61458556

The tuple decreases monotonically from 17.617 (the brick $\pi/2$) to the limit $27/4 = 6.75$ as $k \rightarrow \infty$: the thinner the brick, the closer the system to the baseline. The surviving brick $\pi/7$ occupies an intermediate position on the ladder and is distinguished not by the value of the root but by the spinor sector $bC = \pi^2/98$ and the Berry closure of $\ln(\tau^*)$ with remainder +0.062% (see monographs No. 1 and No. 7).

Layer B4 (the phantom scan) showed that the phantom nearest to the ladder $\pi/30$ shifts under the bricks by at most +1.139% (the best case $\delta = \pi/3$), and on no rung does a phantom approach the ladder within 1% — the ladder creates no spinor clock at the linear level. Layer B5 pinned the sawtooth book defect $F = 26.79$, κ -independent: the defect belongs to the κ -free sector and takes no part in selection.

6. The Tick Obstruction: the Honest Layer B3

A separate part of the campaign measured the tick obstruction — the residual incompatibility of the closure when the linear freedom of the tick is lifted onto the set $\{F = 0, r = 0\}$ — as a function of δ on the same ladder. The result is important on the negative side: the dependence is non-monotone, is not described by a power law (slope in log-log +0.65 at the residual $\ln 2.1$) and reaches its minimum at $k = 12$, not at $k = 7$. The surviving brick $\pi/7$ is therefore not selected by the tick; the relative advantage of one brick over the

baseline (0.032 versus 0.043 at kick 0.05) remains a compatibility anchor, not a selection principle.

This separation — the clock is invariant, the selection is not algebraic — is the substantive core of the theorem: the law (2.1) does not change from brick to brick, and the choice of the concrete value of δ requires a mechanism beyond the algebraic books (finite-amplitude monodromy or Berry holonomy — monograph No. 10).

7. Corollaries

Corollary 2.1 (robustness of the clock pair). The exactness of formula (2.1) on all rungs shows that Theorem 1 is not a property of two special values of κ but a general law of the corrected system; the uniqueness of the common root survives the continuous deformation of the constant along the ladder $\delta_k = \pi/k$.

Corollary 2.2 (the two-point protocol is legitimate). The verification practice of the repository executes the machines at two points ($\kappa = 2$ and $\kappa = 2 - \pi^2/98$). Theorem 2 justifies the protocol: both points lie on one exact curve, so the two-point check fixes both the shape of the curve and its invariant station ladder.

Corollary 2.3 (ticks and selection are separated). Since the tick minimum does not coincide with the surviving brick, the algebraic sector (books + tick) leaves δ free; this is consistent with the zero elimination ideal of the structural barrier (monograph No. 10) and moves the selection question into the transcendental inventory $\{\pi, \ln(16/9)\}$.

8. Honest Caveats

The ladder is discrete: the theorem is proved on twelve rungs, not for all $\delta \in (0, \pi/2)$. The continuous version follows from the structure of the factorizations (the clock factor is quadratic for every δ), but formally the step beyond the ladder is an extrapolation. The form (2.1) uses the additive convention $\kappa(\delta) = 2 - \delta^2/2$; the monograph-sign multiplicative convention writes the same tuple differently and moves $\ln(\tau^*)$ away from the observed value (-7.43%) — the choice of convention remains open. Layer B1 confirms the brick independence of the flat line within the float64 representation; the symbolic version $F \equiv 0$ is verified at the rung points, but not along the whole line as a function of δ . Finally, the

tick is measured at two kick amplitudes (0.05 and 0.01); a finer dependence on the amplitude remains outside the scope of the present theorem.

9. Reproducibility

Table 2.3. Reproduction protocol for Theorem 2

Step	Command	Result
1	<code>python3 brick_scan_tick.py</code>	results/brick_scan_tick.json (layers B1–B5, ladder 12/12)
2	<code>python3 clock_closure_t1c.py</code>	results/clock_closure_t1c.json (two-point control)
3	<code>python3 -m pytest tests/ -q</code>	58 passed (the ladder tests included)

The total running time is minutes on a laptop and comparable in Termux (the machines are single-threaded). The data correspond to commit 5b256cd; an autonomous copy of the machine lives in `einstein_direct/open9/machines/` and runs from its own folder without repository paths.

References

- [1] Repository `wild8highlander/choptuik_ac_bc`: `einstein_direct/brick_scan_tick.py`; `einstein_direct/results/brick_scan_tick.json` (campaign v18a, LEDGER §29).
- [2] Ibid.: `einstein_direct/clock_closure_t1c.py`; `einstein_direct/results/clock_closure_t1c.json` (campaign v17, clock pairs at two points).
- [3] Ibid.: `einstein_direct/s4_berry_one_brick.py` — the Berry screening $\kappa = 2 - \pi^2/98$.
- [4] Ibid.: `einstein_direct/README.md` §§2–3; `einstein_direct/README_EN.md` §29.
- [5] Monograph No. 1 of the theorems12 cycle: the clock-pair theorem.
- [6] M. W. Choptuik. Phys. Rev. Lett. 70, 9 (1993) — source of the comparison anchor $\gamma_{\text{Ch}} = 0.374$.